

Homework 4

1. Find the radius of convergence and the convergence set for each of the following series:

(a) $\sum_{n \geq 0} \frac{nx^n}{2^n}$.

(b) $\sum_{n \geq 1} \frac{x^{2n}}{\sqrt{n}}$.

(c) $\sum_{n \geq 1} (-1)^n \frac{(x-1)^n}{n}$.

2. Study the convergence and compute the sum for the series $\sum_{n \geq 2} \frac{x^n}{n(n-1)}$.

3. For this question you have to use **Python**.

Illustrate that $\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = \ln 2$ by computing partial sums.

Change the order of summation in this series – for example by first adding p positive terms, then q negative terms, and so on – and show numerically that the rearrangement gives a different sum (depending on p, q).